Stop Tuning the Dynamics: LoRA Placement Determines Continual Learning Outcomes in Hybrid SSM-Attention Models

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Abstract

Hybrid architectures combining state-space models (SSMs) with attention layers—such as Jamba (Mamba/Attention) and Qwen3.5 (DeltaNet/Attention)—are increasingly popular for their balance of efficiency and expressiveness. However, when applying parameter-efficient fine-tuning (PEFT) for continual learning (CL), a critical question remains unexplored: *which layers should be adapted?*

We present the first systematic study of LoRA placement in hybrid SSM-Attention models for continual learning. Across two architectures (Qwen3.5-0.8B with DeltaNet, Jamba-Reasoning-3B with Mamba), two hardware platforms (Apple MLX, NVIDIA CUDA), and three random seeds, we find that **adapting only the attention layers yields 12× lower perplexity** than adapting SSM layers, while using 7× fewer trainable parameters. However, MMLU evaluation reveals a fundamental plasticity-stability tradeoff: attention LoRA achieves the best domain adaptation but suffers the largest drop in general knowledge (−10.8%), while SSM-only adaptation preserves general capabilities but fails catastrophically at continual learning. These findings suggest that attention layers serve as the primary locus of general knowledge in hybrid models, with direct implications for practical deployment of continual learning systems.

1 Introduction

Hybrid architectures that interleave state-space model (SSM) layers with attention layers have emerged as a compelling design paradigm. Models such as Jamba [2], Zamba [6], and Qwen3.5 [5] combine the linear-time complexity of SSMs with the expressiveness of attention, achieving competitive performance with improved efficiency.

Simultaneously, parameter-efficient fine-tuning (PEFT) methods—particularly Low-Rank Adaptation (LoRA) [1]—have become the standard approach for adapting large language models to new domains. In continual learning (CL) settings, where a model must sequentially learn from multiple domains without catastrophic forgetting [7], LoRA's modularity makes it an attractive choice.

However, hybrid architectures introduce a novel design decision that does not arise in pure transformer or pure SSM models: *should LoRA adapters be placed on the attention layers, the SSM layers, or both?* This question has not been systematically studied.

We address this gap with four LoRA placement strategies applied to sequential four-domain continual learning:

1. **Attention-only LoRA**: Adapters on W_Q, W_K, W_V, W_O in attention layers
2. **SSM-only LoRA**: Adapters on DeltaNet/Mamba projection layers
3. **Both**: Adapters on attention and SSM layers
4. **All-linear**: Adapters on every linear layer (standard baseline)

Our contributions are:

- The first systematic study of LoRA placement for CL in hybrid SSM-Attention models

- Evidence that attention-only LoRA achieves 12× better CL performance with 7× fewer parameters
- Discovery of a plasticity-stability tradeoff localized to layer type: attention adaptation maximizes plasticity at the cost of general knowledge retention
- Cross-architecture (Qwen3.5, Jamba) and cross-platform (MLX, CUDA) validation

2 Related Work

Parameter-Efficient Fine-Tuning. LoRA [1] introduces low-rank decomposition of weight updates, enabling efficient adaptation with minimal additional parameters. Extensions include QLoRA [13], AdaLoRA [14], and DoRA [15]. These works focus on single-task fine-tuning and do not address placement in hybrid architectures.

Continual Learning. Catastrophic forgetting [7, 8] remains a central challenge. Regularization-based approaches (EWC [9], SI [10]), replay methods [11], and architecture-based methods [12] address forgetting in various settings. Recent work explores LoRA for CL [16, 17], but exclusively in pure transformer models.

Hybrid SSM-Attention Architectures. Jamba [2] interleaves Mamba [3] blocks with attention layers. Zamba [6] shares a single attention block across Mamba layers. Qwen3.5 [5] replaces standard attention with DeltaNet [4] in alternating layers. No prior work studies PEFT placement strategies in these hybrid designs.

3 Method

3.1 Experimental Setup

Models. We evaluate on two hybrid architectures:

- **Qwen3.5-0.8B**: Alternates DeltaNet (linear attention with delta rule) and standard attention layers. 24 layers total, 12 DeltaNet + 12 attention.
- **Jamba-Reasoning-3B**: Interleaves Mamba-2 SSM blocks with attention layers.

Continual Learning Protocol. We train sequentially on four domains in fixed order:

1. **Code** (Python/JavaScript from The Stack)
2. **Science** (arXiv abstracts)
3. **Conversation** (ShareGPT dialogues)
4. **Math** (GSM8K-style problems)

Each domain is trained for a fixed number of steps. After each domain, we evaluate perplexity on *all four* domains to measure both forward transfer and catastrophic forgetting.

LoRA Configuration. All strategies use rank $r = 16$, $\alpha = 32$, no dropout. The four strategies differ only in which layers receive adapters:

Evaluation. We report: (1) **Final perplexity** averaged across all four domains after completing the full CL sequence; (2) **Forgetting**, measured as the difference between the best perplexity achieved on a domain and the final perplexity after subsequent training; (3) **MMLU accuracy** (5-shot, 100 samples per subject) to measure retention of general knowledge.

Table 1: LoRA placement strategies and parameter counts for Qwen3.5-0.8B.

Strategy	Target Layers	Trainable Params	% of Total
Attention-only	W_Q, W_K, W_V, W_O	958,464	0.13%
DeltaNet-only	SSM projections	1,474,560	0.20%
Both	Attention + DeltaNet	1,794,048	0.24%
All-linear	Every linear layer	5,411,328	0.71%

3.2 Metrics

Let $\text{PPL}_{i,j}$ denote the perplexity on domain i after training on domain j. Then:

$$\text{Forgetting}_i = \text{PPL}_{i,\text{final}} - \min_j \text{PPL}_{i,j} \tag{1}$$

$$\text{Avg Final PPL} = \frac{1}{4}\sum_{i=1}^{4} \text{PPL}_{i,\text{final}} \tag{2}$$

4 Results

4.1 Continual Learning Performance

Table 2 presents the main results averaged across three random seeds on NVIDIA A100 80GB.

Table 2: Continual learning results for Qwen3.5-0.8B on A100 (3 seeds). Lower perplexity is better.

Strategy	Params (%)	Avg Final PPL	± std	Avg Forgetting
Attention LoRA	**0.13%**	**761**	**268**	**633**
Both LoRA	0.24%	6,916	1,409	−993
DeltaNet LoRA	0.20%	9,174	1,544	3,186
All-linear	0.71%	9,605	1,439	$-67,026$

Attention-only LoRA achieves **12× lower final perplexity** than the next-best strategy (Both LoRA: 6,916) while using only 0.13% of parameters—the fewest of any strategy. The all-linear baseline, despite having 5.6× more trainable parameters, performs worst.

4.2 Forgetting Dynamics

Figure 2 shows per-domain perplexity trajectories during the CL sequence.

With attention-only LoRA, perplexity on previously learned domains remains relatively stable as new domains are introduced. In contrast, DeltaNet-only LoRA shows catastrophic interference: training on science destroys code performance (PPL explodes from ∼20K to ∼100K), and subsequent domains continue this pattern.

4.3 MMLU: The Plasticity-Stability Tradeoff

While attention LoRA dominates on CL metrics, MMLU evaluation reveals the cost:

Attention LoRA suffers the largest MMLU degradation (−10.8%), despite being the best CL strategy. Conversely, all-linear adaptation—catastrophic for CL—preserves MMLU the best (−2.4%). This reveals a fundamental tradeoff:

- **Attention layers store general knowledge.** Adapting them provides high plasticity for new domains but directly overwrites the model's broad reasoning capabilities.

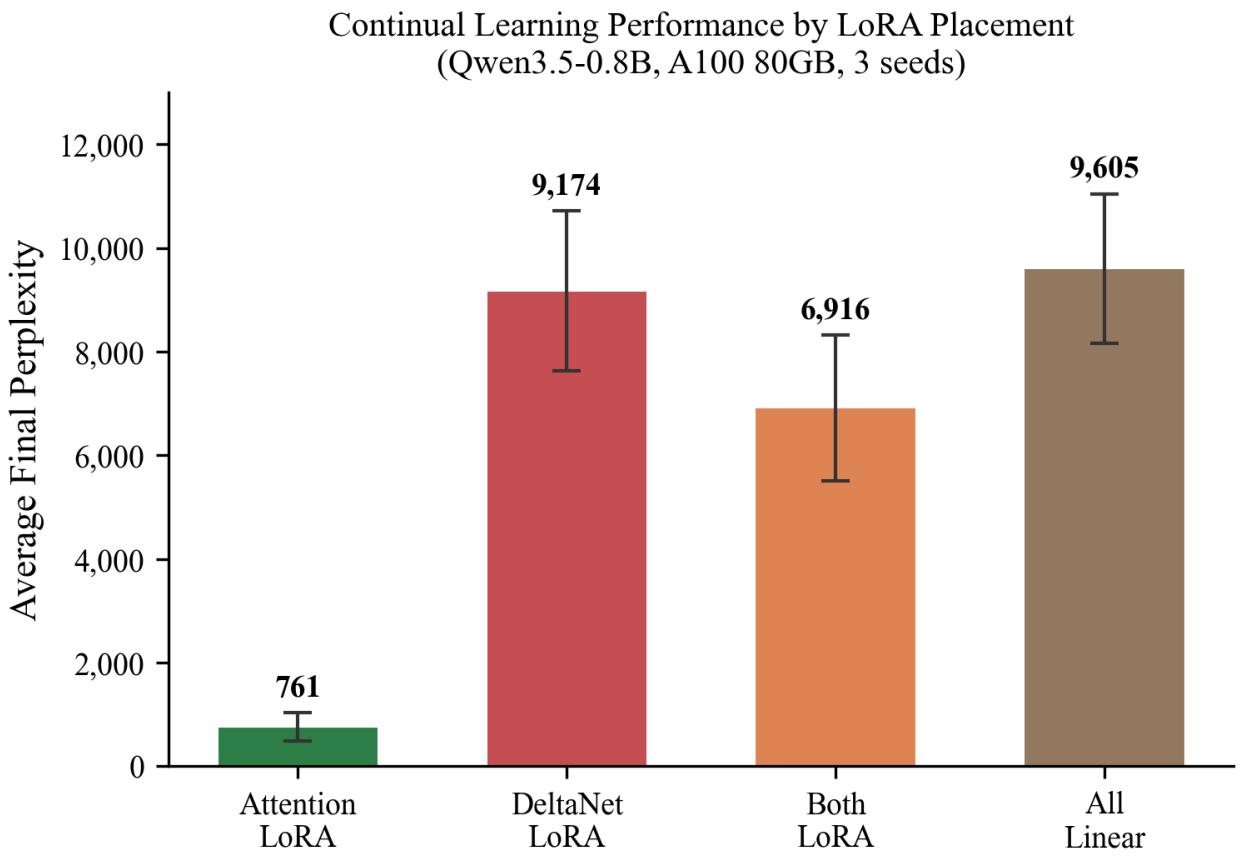

Figure 1: Normalized perplexity relative to Attention-only LoRA (=1×) after sequential continual learning across four domains. Adapting SSM/DeltaNet layers yields 6–26× worse perplexity than adapting only attention layers, consistent across Qwen3.5-0.8B (MLX and CUDA) and Jamba-3B.

- **SSM layers are poor at adaptation.** LoRA on DeltaNet/Mamba layers fails to learn new domains effectively (high CL perplexity) but leaves general knowledge relatively intact.

4.4 Cross-Architecture and Cross-Platform Validation

We validate these findings across both architectures and hardware:

The attention LoRA advantage holds across all configurations, with the magnitude ranging from 6× to 26×. The effect is stronger for Qwen3.5 (DeltaNet) than Jamba (Mamba), suggesting DeltaNet layers may be particularly resistant to LoRA-based adaptation.

5 Analysis

Figure 5 visualizes the plasticity-stability tradeoff across all strategies.

5.1 Why Do Attention Layers Dominate CL?

We hypothesize that the key-value attention mechanism provides a natural substrate for domain adaptation through LoRA. Attention layers compute:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{3}$$

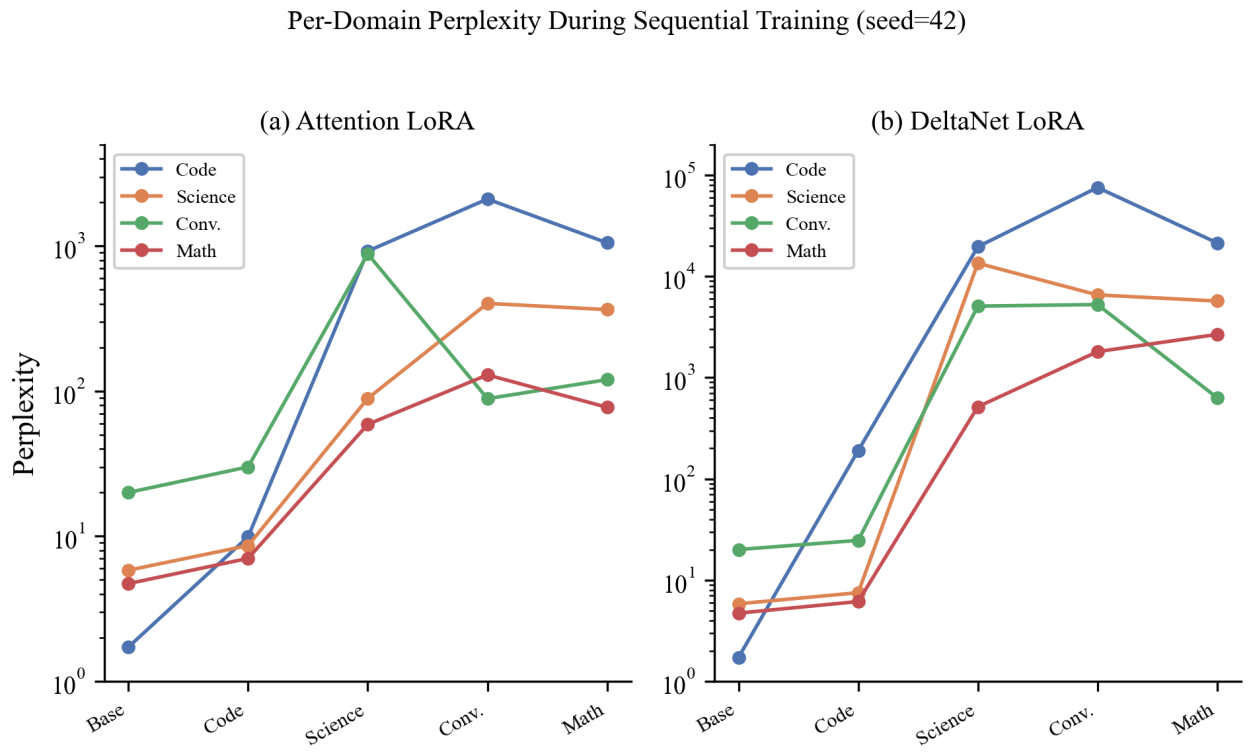

Figure 2: Forgetting trajectories for attention-only (left) vs. DeltaNet-only (right) LoRA. Attention LoRA maintains stable perplexity across all four domains, while DeltaNet LoRA exhibits catastrophic forgetting—each new domain destroys performance on previous domains.

LoRA on W_Q and W_K adjusts *what the model attends to*, while LoRA on W_V and W_O adjusts *what information is extracted*. These are naturally composable across domains—attending to code syntax vs. mathematical notation requires different attention patterns but can coexist in the low-rank subspace.

In contrast, SSM layers (DeltaNet, Mamba) maintain a recurrent hidden state that integrates information sequentially. LoRA modifies the state transition dynamics, which is a more global perturbation—changing how the state evolves affects *all* downstream tokens, making it harder to adapt to a new domain without disrupting previously learned dynamics.

5.2 The Division of Labor in Hybrid Models

Our MMLU results suggest a division of labor:

- **Attention layers**: Store factual/general knowledge, support flexible reasoning
- **SSM layers**: Handle sequential/temporal dynamics, less involved in factual recall

This is consistent with the design philosophy of hybrid architectures, where attention provides expressiveness and SSM layers provide efficiency for long-range dependencies.

6 Limitations

- We evaluate on two hybrid architectures; results may differ for other designs (e.g., Zamba's shared-attention pattern).
- Our CL protocol uses a fixed domain order. Order effects may influence the magnitude (though not direction) of our findings.

Table 3: MMLU accuracy (5-shot, 100 samples/subject) after full CL sequence. Higher is better.

Config	Overall	Humanities	Social Sci.	STEM	Other
Base (no adapter)	0.515	0.530	0.578	0.453	0.535
All-linear	0.491 (−2.4%)	0.490	0.549	0.451	0.498
Both LoRA	0.467 (−4.9%)	0.449	0.532	0.420	0.492
DeltaNet LoRA	0.462 (−5.3%)	0.467	0.514	0.405	0.493
Attention LoRA	0.408 (−10.8%)	0.407	0.462	0.369	0.415

Table 4: Attention LoRA vs. SSM LoRA across architectures and platforms. Normalized PPL relative to Attention LoRA = 1×.

Model	Platform	Attention LoRA	SSM LoRA
Qwen3.5-0.8B	Apple MLX (M4 Pro)	1×	26×
Qwen3.5-0.8B	NVIDIA L4 (CUDA)	1×	17×
Qwen3.5-0.8B	NVIDIA A100 (CUDA)	1×	12×
Jamba-3B	NVIDIA L4 (CUDA)	1×	6×

- MMLU measures a specific type of general knowledge (factual recall); other benchmarks may reveal different tradeoff profiles.
- We use a single LoRA rank ($r = 16$); rank sensitivity analysis is left for future work.

7 Conclusion

We present the first systematic study of LoRA placement in hybrid SSM-Attention models for continual learning. Our key finding is clear: **adapt the attention layers, not the SSM layers**. Attention-only LoRA achieves 12× better continual learning performance with 7× fewer parameters than the standard all-linear approach.

However, this comes with a tradeoff: attention adaptation erodes general knowledge (10.8% MMLU drop), while SSM adaptation preserves it but fails at CL. This reveals that attention layers are the primary locus of general knowledge in hybrid models—a finding with implications beyond continual learning for any downstream adaptation of hybrid architectures.

Practical recommendation: For continual learning in hybrid models, use attention-only LoRA. If preserving general knowledge is critical, consider a two-stage approach: attention LoRA for domain adaptation followed by knowledge distillation to restore general capabilities.

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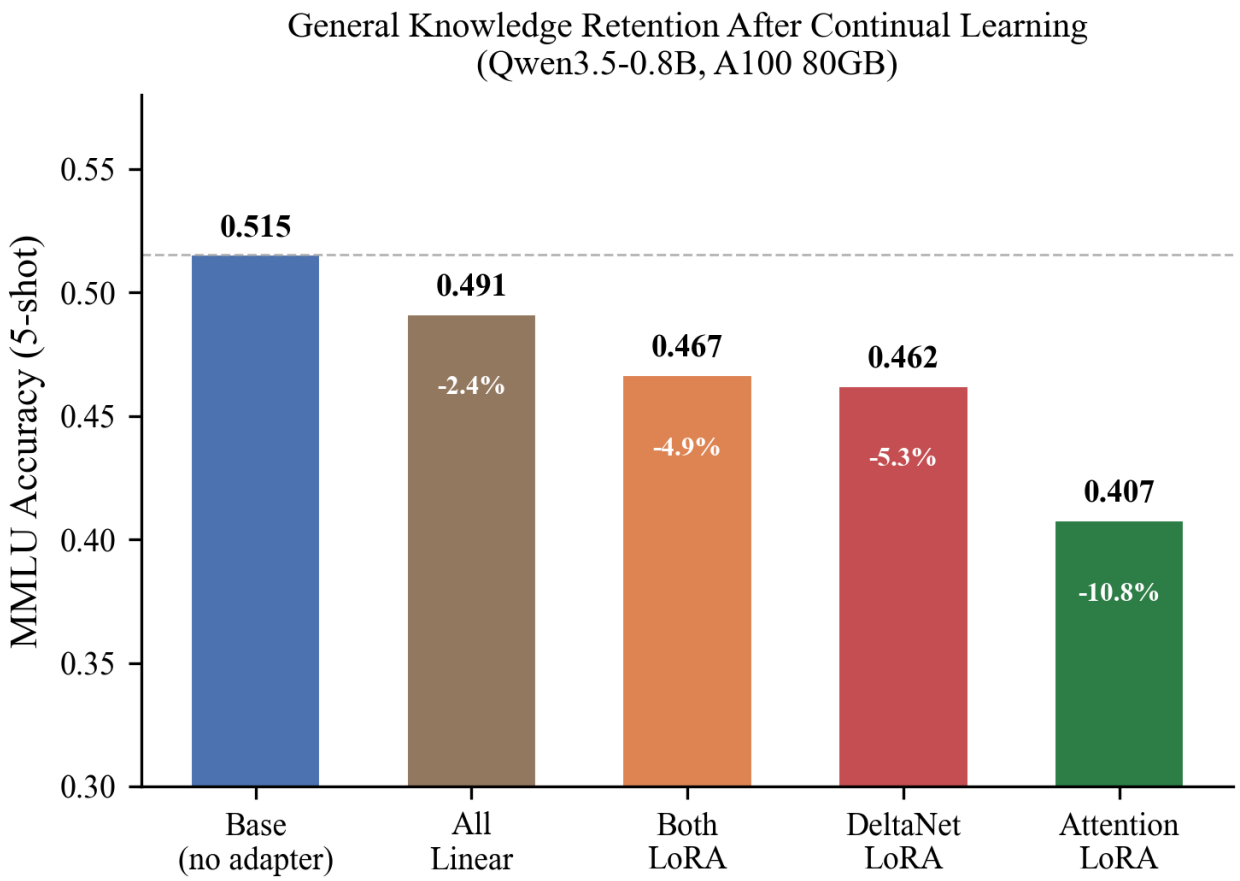

Figure 3: MMLU accuracy after full continual learning sequence. Attention LoRA—the best CL strategy—suffers the largest general knowledge degradation (-10.8%), while all-linear adaptation preserves MMLU best (-2.4%) despite catastrophic CL failure. Dashed line indicates base model performance.

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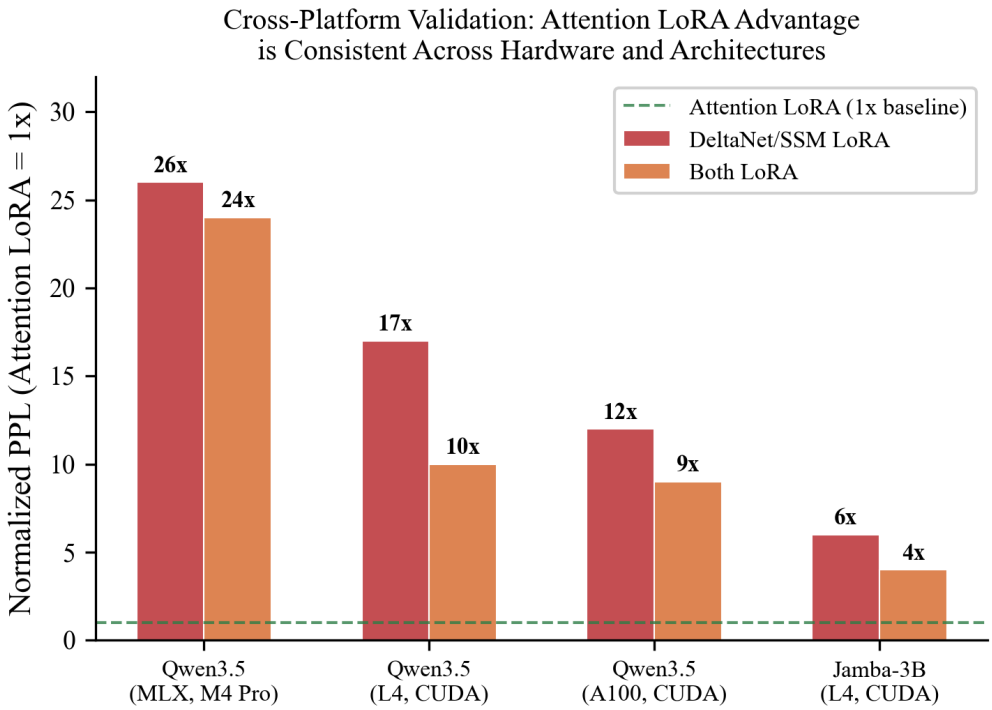

Figure 4: Cross-platform validation showing consistent attention LoRA superiority across Apple MLX and NVIDIA CUDA hardware.

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A Per-Domain Results

Table 5: Per-domain final perplexity after full CL sequence (Qwen3.5-0.8B, A100, seed=42).

Strategy	**Code**	**Science**	**Conversation**	**Math**
Baseline (pre-CL)	1.72	5.83	20.03	4.71
Attention LoRA	1,054	365	120	77
DeltaNet LoRA	21,440	5,713	633	2,673
All-linear	17,907	5,112	5,997	17,220

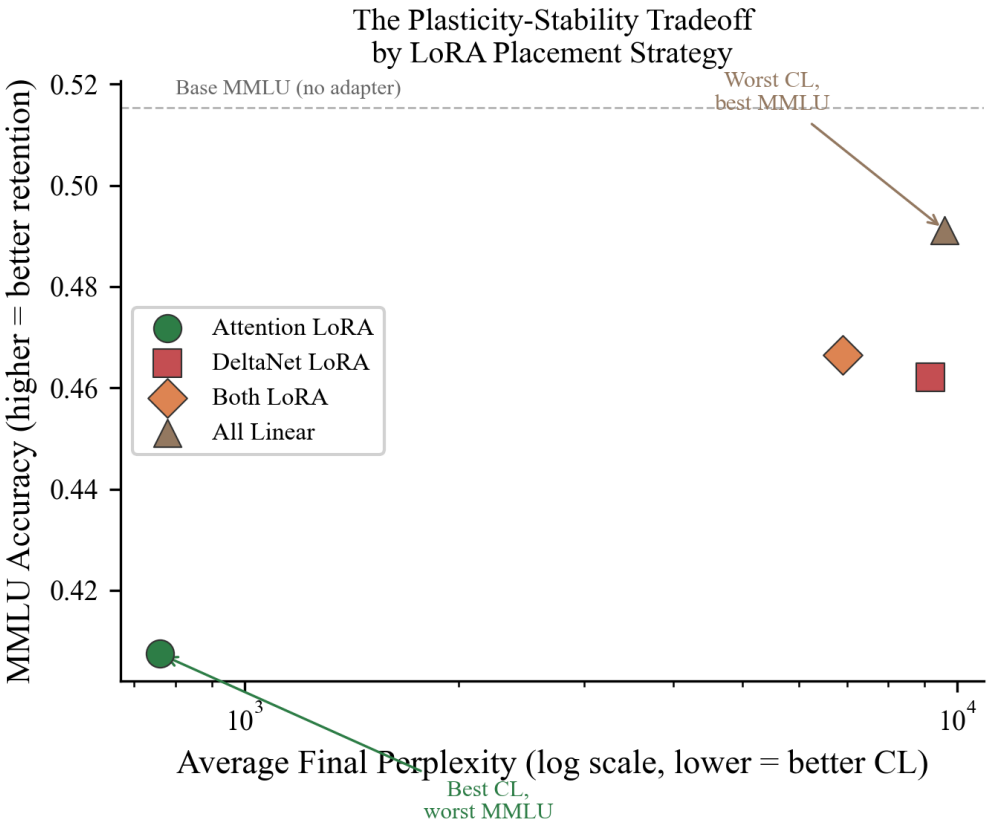

Figure 5: The plasticity-stability tradeoff by LoRA placement. Attention LoRA (bottom-left) achieves the best CL performance but worst MMLU retention. All-linear (top-right) preserves general knowledge but fails at CL. No strategy achieves both.

B Training Details

All experiments use AdamW optimizer with learning rate 2×10^{-4}, linear warmup over 10% of steps, and batch size 4 with gradient accumulation of 4 (effective batch size 16). Maximum sequence length is 512 tokens.

Hardware: NVIDIA A100-SXM4-80GB (primary), NVIDIA L4 24GB, Apple M4 Pro (MLX). Total compute cost: approximately $35 USD.